

DIVYANSHU PATEL

Chennai, India

✉ divyanshupatel.dev@gmail.com

☎ +91 9301503581

🌐 divyanshupatel.com

🌐 linkedin.com/in/patel-divyanshu

Summary

Computer Science undergraduate specializing in AI and ML with hands-on experience in production mobile applications, deep learning systems, and space-tech research. Currently a Summer Trainee at URSC ISRO, Bengaluru, applying Graph Neural Networks to satellite constellation routing. Independent developer of a Flutter app serving 4,000+ users.

Education

Vellore Institute of Technology (VIT), Chennai
B.Tech in Computer Science and Engineering (AI and ML)

Aug 2023 – May 2027
CGPA: 8.78 / 10

Experience

Summer Research Trainee

May 2025 – Present

URSC – ISRO, Bengaluru — MPOG Department

- Applying Graph Neural Networks to optimize inter-satellite routing in LEO constellations, modelling topology as dynamic time-varying graphs to minimize latency, link contention, and handover overhead.
- Benchmarking GNN-based routing against Dijkstra, shortest-path, and flow-based baselines across simulated multi-plane constellation scenarios.
- Aligning algorithmic outputs with MPOG operational constraints and real-world satellite scheduling requirements in collaboration with URSC scientists.

Independent Software Developer

Jan 2024 – Feb 2026

VIT Verse – Student Productivity Mobile Application

- Designed, built, and shipped a production Flutter app serving 4,000+ active users, owning the full lifecycle from system design and UI/UX to deployment, monitoring, and iteration.
- Implemented offline-first architecture with Firebase, Supabase, SQLite, and REST APIs; integrated push notifications, real-time sync, and secure media storage.
- Optimized performance via caching, lazy loading, and widget-level rebuilds; monitored crash analytics to drive iterative improvements over two academic years.

Projects

Self-Healing Monocular Digital Twin System

GitHub

- Built an end-to-end monocular digital twin pipeline for autonomous edge vehicles – ingesting dashcam video, simulating adverse conditions (fog, blur, occlusion), and recovering degraded regions via CLAHE-based contrast restoration and selective sharpening.
- Integrated YOLOv8 object detection with monocular 2D-to-3D spatial mapping using scale heuristics to reconstruct approximate scene geometry from a single RGB camera with no depth sensor.
- Visualized reconstructed Gaussian-splatting-inspired scenes in a live React + Three.js dashboard with real-time WebSocket-streamed telemetry, backed by a FastAPI vision processing server.

Leukemia Diagnosis System

GitHub

- Developed an end-to-end deep learning pipeline for multi-class leukemia cell classification using EfficientNetV2 trained on 3,256 labeled blood-cell images, achieving 98.93% accuracy.
- Applied transfer learning with fine-tuning and data augmentation to overcome limited medical dataset constraints and improve cross-class generalization across distinct cell morphologies.
- Implemented Grad-CAM heatmaps for clinically interpretable predictions and evaluated model robustness with precision, recall, F1-score, and per-class confusion matrices.

Activities & Achievements

Competed in 20+ hackathons, building and shipping rapid prototypes across domains including AI, mobile, and systems engineering. Gained hands-on experience in fast iteration, cross-functional collaboration, and delivering MVPs under tight deadlines.

Technical Skills

Languages: C++, Java, Python, JavaScript, Dart

Frameworks: Flutter, React, TensorFlow, PyTorch, scikit-learn, Streamlit

Cloud & Backend: Firebase, Supabase, Google Cloud Storage, REST APIs, SQL, SQLite

Core Areas: Mobile Development, Graph Neural Networks, Deep Learning, Explainable AI, System Design, Full-Stack Systems